WORKSHEET

Chapter 9: Drawing directed graphs

1 Draw the appropriate directed graph for each of the following activity tables.

a

Activity	Time	Immediate predecessor/s
A	2	None
B	3	*A*
C	3	None
D	4	*C*
E	1	*B*, *D*

b

Activity	Time	Immediate predecessor/s
A	8	None
B	1	None
C	2	*B*
D	4	*C*
E	2	*A*, *D*

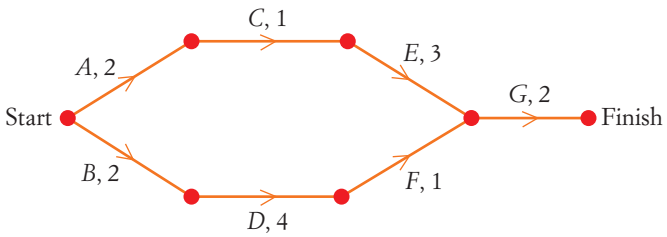
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c

Activity	Time	Immediate predecessor/s
A	2	None
B	1	*A*
C	3	*A*
D	4	*B*
E	3	*C*
F	6	*D*, *E*

2 Complete the activity table for each of the following directed graphs.

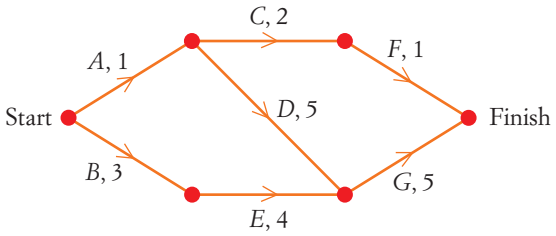
a



Activity	Time	Immediate predecessor/s
A		
B		
C		
D		
E		
F		
G		

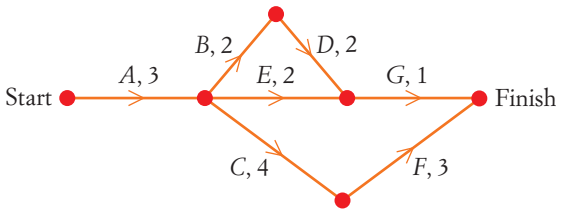
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b



Activity	Time	Immediate predecessor/s
A		
B		
C		
D		
E		
F		
G		

c



Activity	Time	Immediate predecessor/s
A		
B		
C		
D		
E		
F		
G		

3 Draw the network for each of the following activity tables, including dummy activities where required.

a

Activity	Time	Immediate predecessor/s
A	1	None
B	2	None
C	1	None
D	1	*A*, *B*, *C*

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b

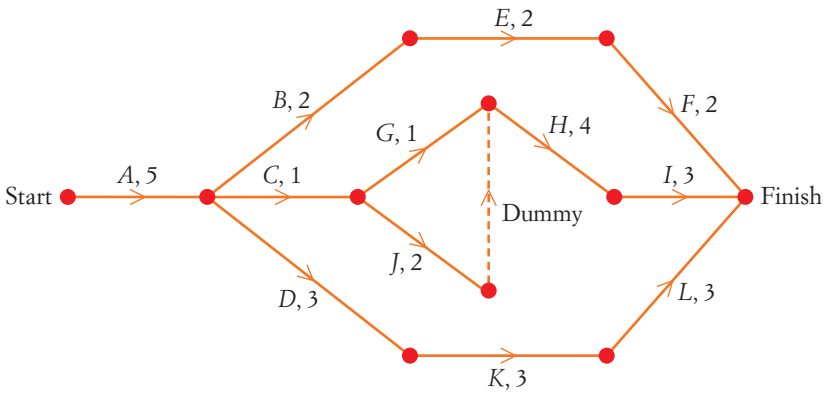
Activity	Time	Immediate predecessor/s
A	2	None
B	1	None
C	4	*A*, *B*
D	2	*C*
E	2	*C*
F	2	*D*, *E*

c

Activity	Time	Immediate predecessor/s
A	2	None
B	1	*A*
C	4	None
D	2	*C*
E	1	*D*
F	1	*E*, *I*
G	2	*B*, *F*
H	3	*C*
I	2	*H*

4 Complete the activity table for each of the following networks.

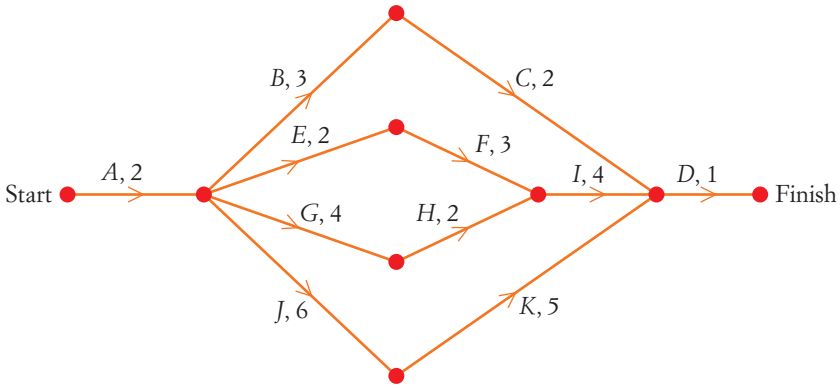
a



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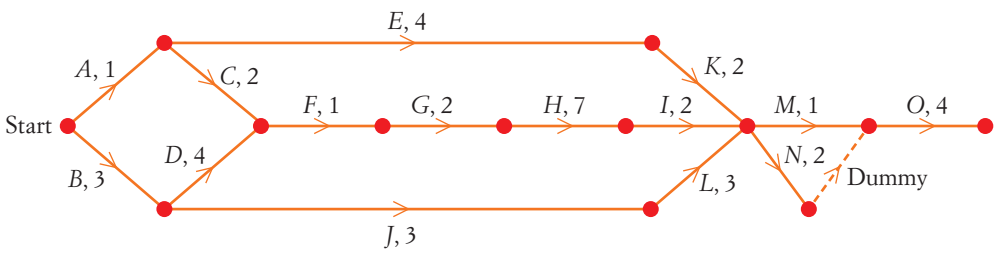
Activity	Time	Immediate predecessor/s
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		
K		
L		

b



Activity	Time	Immediate predecessor/s
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		
K		

c



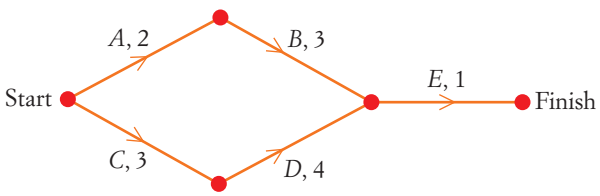
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Activity	Time	Immediate predecessor/s
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		
K		
L		
M		
N		
O		

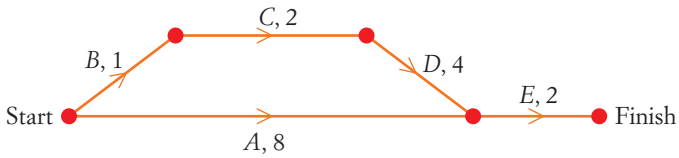
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Answers

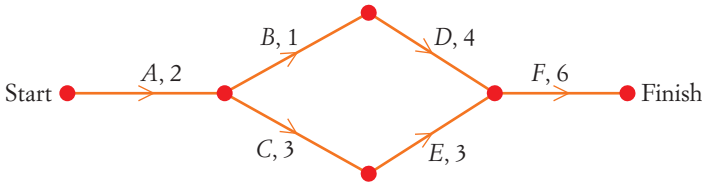
1 a



b



c



2 a

Activity	Time	Immediate predecessor/s
A	2	None
B	2	None
C	1	*A*
D	4	*B*
E	3	*C*
F	1	*D*
G	2	*E*, *F*

b

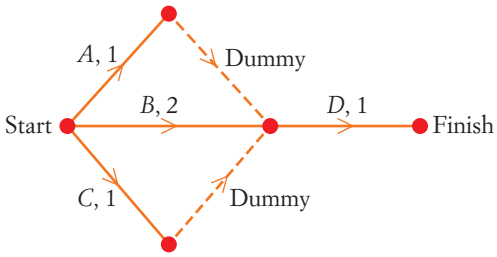
Activity	Time	Immediate predecessor/s
A	1	None
B	3	None
C	2	*A*
D	5	*A*
E	4	*B*
F	1	*C*
G	5	*D*, *E*

c

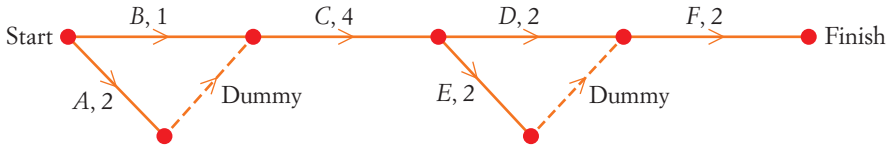
Activity	Time	Immediate predecessor/s
A	3	None
B	2	*A*
C	4	*A*
D	2	*B*
E	2	*B*
F	3	*C*
G	1	*D*, *E*

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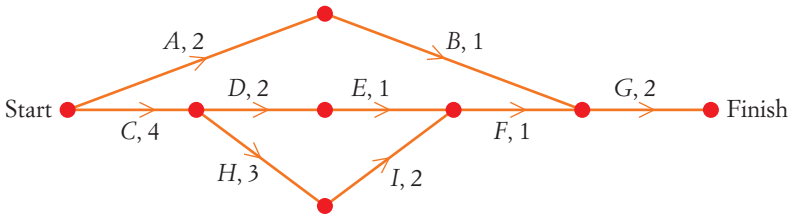
3 a



b



c



4 a

Activity	Time	Immediate predecessor/s
A	5	None
B	2	*A*
C	1	*A*
D	3	*A*
E	2	*B*
F	2	*E*
G	1	*C*
H	4	*G, J*
I	3	*H*
J	2	*C*
K	3	*D*
L	3	*K*

b

Activity	Time	Immediate predecessor/s
A	2	None
B	3	*A*
C	2	*B*
D	1	*C, I, K*
E	2	*A*
F	3	*E*
G	4	*A*
H	2	*G*
I	4	*F, H*
J	6	*A*
K	5	*J*

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c

Activity	Time	Immediate predecessor/s
A	1	None
B	3	None
C	2	*A*
D	4	*B*
E	4	*A*
F	1	*C*, *D*
G	2	*F*
H	7	*G*
I	2	*H*
J	3	*B*
K	2	*E*
L	3	*J*
M	1	*K*, *I*, *L*
N	2	*K*, *I*, *L*
O	4	*M*, *N*

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